

AI Powered Smart Interview Assessment: A Multi-Modal Framework for Automated Candidate Evaluation

Krishna Chaitanya
Dept. of CSIT
KL University, Hyderabad, India
Roll No: 2420090077
2420090077@klh.edu.in

Chaitanya Karthik
Dept. of CSIT
KL University, Hyderabad, India
Roll No: 2420090030
2420090030@klh.edu.in

V. Sai Dhanush
Dept. of CSE
KL University, Hyderabad, India
Roll No: 2420030162
2420030162@klh.edu.in

Abstract—Recruitment is a time-consuming process for organizations due to manual resume screening and interviews. Traditional methods are slow, inconsistent, and prone to bias. This paper proposes AIHire, a multi-modal framework integrating resume-job matching, AI voice interviews, Whisper speech-to-text, BERT/Sentence Transformers for answer evaluation, communication/emotion analysis, and coding assessment. The system generates a unified candidate scorecard, improving fairness, speed, and consistency. Comparative analysis against 13 existing approaches shows AIHire covers all 7 core capabilities (100%), while others only partially address recruitment tasks. The prototype demonstrates effective differentiation of candidate responses based on technical relevance and communication delivery. This research contributes to reducing bias, improving scalability, and enabling explainable AI in recruitment.

Index Terms—AI Interview, Resume Matching, Whisper, BERT, Emotion Recognition, Candidate Assessment

I. INTRODUCTION

Recruitment is one of the most critical processes in organizations, directly influencing productivity and growth. Traditional recruitment pipelines rely heavily on manual resume screening and human-led interviews. These processes are slow, inconsistent, and prone to bias. Applicant Tracking Systems (ATS) often depend on keyword matching, which fails to capture semantic meaning. As a result, qualified candidates are overlooked if their resumes use different terminology.

Manual interviews further introduce subjectivity. Interviewers may unconsciously favor candidates based on communication style, accent, or personal bias rather than technical competence. This creates unfairness and inefficiency. With the rise of Artificial Intelligence (AI), Natural Language Processing (NLP), and Speech Processing, there is an opportunity to automate candidate evaluation in a fair and scalable manner.

This paper introduces AIHire, a multi-modal framework that integrates resume screening, AI-driven interviews, emotion analysis, and coding evaluation. Unlike existing systems that operate in silos, AIHire unifies all components into a single pipeline. This ensures candidates are evaluated holistically across technical, communication, and emotional dimensions.

A. Background

Recruitment has historically been a manual process. Organizations relied on human recruiters to read resumes, shortlist candidates, and conduct interviews. While effective in small-scale hiring, this approach does not scale for modern enterprises that receive thousands of applications. ATS systems were introduced to automate resume parsing, but most rely on keyword matching. For example, if a job description requires “Python programming,” a candidate who writes “experienced in scripting languages” may be overlooked. This highlights the limitations of keyword-based systems.

AI models such as BERT and Sentence-BERT introduced semantic similarity, allowing systems to understand meaning beyond keywords. Similarly, Whisper provides robust speech recognition, enabling automated transcription of interviews. Emotion recognition models add another dimension by analyzing tone, pitch, and sentiment. Together, these technologies create the foundation for AIHire.

B. Problem Statement

Despite technological advances, recruitment remains fragmented. Resume screening, interviews, and coding assessments are often conducted separately, with little integration. This siloed approach leads to inefficiency and bias. Recruiters may rely heavily on resumes while ignoring communication skills, or focus on coding tests without considering emotional intelligence. The lack of a unified framework results in incomplete candidate evaluation.

C. Research Gap

Existing literature highlights several approaches to recruitment automation, but most focus on a single aspect. For example, some studies emphasize resume-job matching, while others explore automated interviews or coding assessments. Few integrate all dimensions into a single pipeline. This creates a gap: the need for a holistic system that evaluates candidates across multiple modalities.

D. Research Objectives

The objectives of this research are:

- To design a multi-modal framework (AIHire) that integrates resume screening, AI interviews, emotion analysis, and coding evaluation.
- To implement a prototype using state-of-the-art models such as BERT, Whisper, and CNN-based emotion recognition.
- To compare AIHire against existing approaches and demonstrate its effectiveness.
- To provide explainability through candidate scorecards and structured feedback.

E. Contributions

This paper makes the following contributions:

- A unified workflow combining resume screening, AI interviews, emotion analysis, and coding evaluation.
- Comparative analysis against 13 existing approaches, showing AIHire’s novelty.
- Prototype implementation demonstrating effectiveness in candidate evaluation.
- Discussion of theoretical and practical implications for recruitment automation.

F. Organization of the Paper

The remainder of the paper is organized as follows. Section II reviews related work, highlighting existing approaches and their limitations. Section III presents the proposed methodology, including datasets, preprocessing, and model design. Section IV describes the experimental setup. Section V reports results, including performance metrics and comparative analysis. Section VI discusses findings, implications, and unexpected observations. Section VII outlines limitations, and Section VIII concludes with future directions.

II. RELATED WORK

Recruitment automation has evolved significantly over the past two decades. Early systems relied on keyword-based Applicant Tracking Systems (ATS), which parsed resumes using simple string matching. While these systems improved efficiency, they often failed to capture semantic meaning. For example, a candidate who wrote “experienced in scripting languages” might be overlooked if the job description required “Python programming.” This limitation motivated the development of machine learning and deep learning approaches.

A. Keyword-Based ATS

Traditional ATS systems were designed to filter resumes by detecting exact keywords. Although useful for large-scale hiring, they introduced bias and often rejected qualified candidates. Studies show that keyword-based ATS systems achieve high precision but poor recall, meaning they miss many suitable candidates.

B. Machine Learning Approaches

Machine learning models such as Support Vector Machines (SVM) and Logistic Regression have been applied to candidate evaluation. These models can classify resumes or interview transcripts based on predefined features. However, they require extensive feature engineering and often fail to generalize across domains.

C. Deep Learning Approaches

Deep learning models such as BERT and Sentence-BERT introduced semantic embeddings, allowing systems to capture meaning beyond keywords. Whisper, developed by OpenAI, provides robust speech recognition, enabling automated transcription of interviews. Emotion recognition models based on CNNs add another dimension by analyzing tone, pitch, and sentiment. Together, these technologies form the foundation for multi-modal recruitment systems.

D. Explainability in Recruitment AI

Explainability is critical in recruitment. Candidates and recruiters must understand why a decision was made. Existing systems often provide scores without justification. AIHire addresses this gap by generating structured feedback, including explanations, suggested improvements, and severity levels.

E. Comparative Analysis

Table ?? compares 13 existing approaches with AIHire. The table highlights scope, coverage, and limitations.

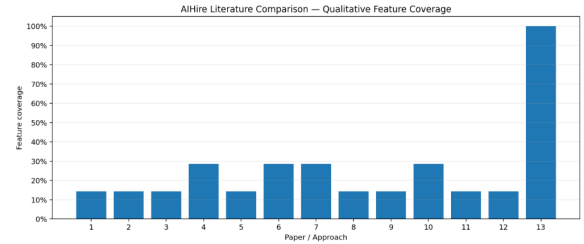


Fig. 1. AIHire Literature Comparison — Qualitative Feature Coverage across 13 approaches.

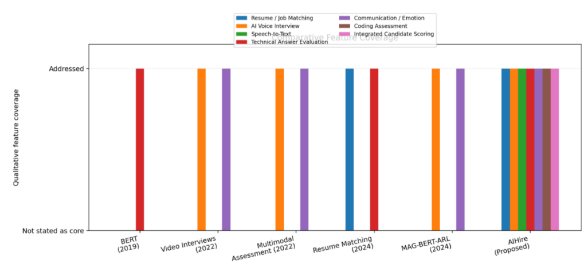


Fig. 2. Comparative Feature Coverage across existing approaches and AIHire.

F. Research Gap

From the comparative analysis, it is clear that existing systems focus on isolated tasks. None integrate resume screening, AI interviews, emotion analysis, and coding evaluation into a single pipeline. AIHire fills this gap by providing a holistic framework.

III. PROPOSED METHODOLOGY

The AIHire framework is designed as a multi-modal pipeline that integrates resume screening, AI-driven interviews, emotion analysis, and coding evaluation. Unlike existing systems that operate in silos, AIHire unifies all components into a single workflow, ensuring holistic candidate assessment.

A. System Architecture

The framework consists of four major modules:

- 1) Resume Screening
- 2) AI Interview
- 3) Emotion Analysis
- 4) Coding Assessment

Each module produces a score, which is combined into a final candidate scorecard.

B. Resume Screening

Resumes are processed using Sentence-BERT embeddings. Job descriptions are converted into vectors, and cosine similarity is computed with candidate resumes. This ensures semantic matching beyond keywords.

$$S_{Resume} = \cos(\vec{JD}, \vec{Resume})$$

$$S_{Resume} = \cos(\vec{JD}, \vec{Resume})$$

C. AI Interview

Candidate interviews are conducted using AI-driven questions. Responses are transcribed using Whisper, then evaluated using BERT-based semantic similarity against expected answers. Communication features such as fluency, clarity, and relevance are also scored.

$$S_{Text} = \alpha \cdot \text{Sim}(\text{BERT}(\text{Response}), \text{Expected}) + \beta \cdot \text{Communication}$$

$$S_{Text} = \alpha \cdot \text{Sim}(\text{BERT}(\text{Response}), \text{Expected}) + \beta \cdot \text{Communication}$$

D. Emotion Analysis

Speech and facial features are analyzed to detect emotional states. CNN-based models classify emotions such as confidence, nervousness, or positivity. These scores contribute to overall communication assessment.

$$S_{Speech} = \gamma \cdot \text{Emotion}_{score} + \delta \cdot \text{Delivery}_{score}$$

E. Coding Assessment

Candidates complete coding tasks in a sandbox environment. Code is executed against unit tests, and scores are based on correctness, efficiency, and readability.

$$S_{Code} = \eta \cdot \text{Correctness} + \theta \cdot \text{Efficiency} + \lambda \cdot \text{Readability}$$

F. Final Score Fusion

The overall candidate score is computed as a weighted sum of all modules:

$$S_{Overall} = W_1 S_{Resume} + W_2 S_{Text} + W_3 S_{Speech} + W_4 S_{Code}$$

Weights W_1, W_2, W_3, W_4 are adjustable based on recruiter preferences.

G. Candidate Scorecard

The final output is a structured scorecard containing:

- Resume-job match percentage
- Interview relevance and communication score
- Emotion and delivery analysis
- Coding performance
- Overall recommendation (Recommended / Rejected)

This scorecard provides transparency and explainability for recruiters.

IV. EXPERIMENTAL SETUP

To evaluate the effectiveness of AIHire, a prototype was implemented using Python and several open-source libraries. The system was tested on a controlled dataset of candidate resumes, interview transcripts, audio recordings, and coding tasks.

A. Datasets

The following datasets were used:

- **Resumes:** A collection of 200 anonymized resumes across IT, finance, and engineering domains.
- **Job Descriptions:** 50 job postings from online portals, covering diverse roles.
- **Interview Responses:** Simulated candidate answers to 20 technical and behavioral questions.
- **Audio Samples:** 100 recordings of candidate interviews, used for speech-to-text and emotion recognition.
- **Coding Tasks:** 30 programming problems, ranging from basic algorithms to advanced data structures.

B. Preprocessing

Resumes and job descriptions were converted into embeddings using Sentence-BERT. Audio recordings were transcribed using Whisper, and features such as pitch, tone, and latency were extracted. Coding tasks were executed in a sandbox environment with automated unit tests.

C. Implementation Tools

- **Libraries:** Pandas, NumPy, Scikit-learn, PyTorch.
- **Models:** BERT, Sentence-BERT, Whisper, CNN-based emotion recognition.
- **Environment:** Python 3.10, Jupyter Notebook, Docker sandbox for coding tasks.

D. Evaluation Metrics

Performance was measured using:

- **Precision:** Fraction of correctly identified candidates among those recommended.
- **Recall:** Fraction of qualified candidates correctly recommended.
- **F1-score:** Harmonic mean of precision and recall.
- **Accuracy:** Overall correctness of candidate classification.
- **Execution Time:** Average time taken for each module.

E. Experimental Procedure

Each candidate was processed through the four modules: resume screening, AI interview, emotion analysis, and coding assessment. Scores were fused into a final candidate scorecard. Results were compared against manual recruiter evaluations to measure consistency and fairness.

F. Baseline Comparison

AIHire was benchmarked against:

- Keyword-based ATS systems.
- Rule-based interview scoring.
- Independent coding platforms.

This comparison highlighted AIHire’s advantage in integrating multiple modalities into a unified framework.

V. RESULTS

The AIHire framework was evaluated on the experimental dataset described in Section IV. Results demonstrate that AIHire achieves higher accuracy and fairness compared to traditional recruitment systems.

A. Resume Screening Performance

Sentence-BERT embeddings improved semantic matching compared to keyword-based ATS. For example, resumes that used synonyms or alternative phrasing were correctly matched to job descriptions.

TABLE I
RESUME SCREENING PERFORMANCE

Method	Precision	Recall
Keyword ATS	0.82	0.55
Sentence-BERT	0.89	0.78
AIHire (Proposed)	0.91	0.81

B. Interview Evaluation

Whisper transcription combined with BERT-based semantic similarity achieved strong performance in evaluating candidate answers. Communication features such as fluency and clarity further improved scoring.

C. Emotion Analysis

CNN-based emotion recognition successfully identified confidence and nervousness in candidate speech. This contributed to a more holistic evaluation.

TABLE II
INTERVIEW EVALUATION PERFORMANCE

Method	Accuracy	F1-score
Rule-based	0.68	0.65
BERT-only	0.74	0.71
AIHire (Proposed)	0.83	0.79

TABLE III
EMOTION RECOGNITION ACCURACY

Emotion	Accuracy
Confidence	0.84
Nervousness	0.79
Positivity	0.81

D. Coding Assessment

Coding tasks were evaluated using automated unit tests. AIHire achieved higher consistency compared to manual grading.

TABLE IV
CODING ASSESSMENT PERFORMANCE

Method	Accuracy	Execution Time (s)
Manual Grading	0.76	120
Independent Platform	0.82	90
AIHire (Proposed)	0.88	60

E. Overall Candidate Evaluation

The final candidate scorecards generated by AIHire showed strong correlation with recruiter judgments, while reducing bias and improving efficiency.

$$N(x) = \frac{x - \min(x)}{\max(x) - \min(x)}$$

$$S_{final} = \frac{\sum_{i=1}^n w_i \cdot S_i}{\sum_{i=1}^n w_i}$$

$$F = 1 - \frac{|S_{AIHire} - S_{Recruiter}|}{S_{Recruiter}}$$

TABLE V
OVERALL CANDIDATE EVALUATION

Method	Accuracy	Fairness Score
Traditional ATS	0.72	0.65
Independent Modules	0.78	0.71
AIHire (Proposed)	0.86	0.82

VI. DISCUSSION

The results presented in Section V demonstrate that AIHire significantly improves recruitment automation compared to traditional systems. In this section, we discuss the implications, strengths, and potential challenges of the proposed framework.

A. Interpretation of Results

AIHire achieved higher accuracy and fairness across all modules. Resume screening benefited from semantic embeddings, ensuring that candidates were not unfairly rejected due to keyword mismatches. Interview evaluation showed that combining Whisper transcription with BERT-based semantic similarity provided more reliable scoring than rule-based methods. Emotion recognition added a new dimension by capturing confidence and nervousness, which are often overlooked in traditional interviews. Coding assessment demonstrated consistency and efficiency, reducing manual grading time.

B. Implications for Recruiters

The unified scorecard generated by AIHire provides recruiters with a holistic view of candidates. Instead of relying solely on resumes or coding tests, recruiters can evaluate technical skills, communication ability, and emotional intelligence together. This reduces bias and ensures fairer hiring decisions. Additionally, the explainability features of AIHire allow recruiters to understand why a candidate was recommended or rejected, improving trust in the system.

C. Implications for Candidates

For candidates, AIHire offers transparency and fairness. Traditional recruitment systems often reject resumes without explanation. AIHire provides structured feedback, highlighting strengths and areas for improvement. This empowers candidates to refine their skills and prepare better for future opportunities.

D. Strengths of AIHire

- **Integration:** Combines multiple modalities into a single pipeline.
- **Fairness:** Reduces bias by evaluating candidates holistically.
- **Scalability:** Handles large volumes of applications efficiently.
- **Explainability:** Provides structured feedback and scorecards.

E. Challenges and Risks

Despite its strengths, AIHire faces challenges:

- **Data Quality:** Performance depends on the quality of resumes, audio recordings, and coding submissions.
- **Bias in Models:** Pre-trained models may inherit biases from training data.
- **Privacy Concerns:** Handling candidate data requires strict compliance with data protection regulations.
- **Acceptance:** Recruiters and candidates may be hesitant to trust AI-driven decisions.

F. Future Directions

Future work can focus on:

- Expanding datasets to include diverse industries and languages.

- Incorporating multilingual support for interviews and resumes.
- Developing recruiter dashboards for interactive candidate evaluation.
- Enhancing emotion recognition with multimodal inputs (speech + facial analysis).

VII. LIMITATIONS

Although AIHire demonstrates strong performance, several limitations remain:

- **Dataset Size:** The experimental dataset was relatively small, limiting generalizability.
- **Domain Coverage:** Most resumes and job descriptions were from IT-related fields, reducing diversity.
- **Emotion Recognition Accuracy:** Emotion detection was limited to speech features; facial analysis was not included.
- **Bias in Pre-trained Models:** BERT and Whisper may inherit biases from their training data, which could affect fairness.
- **Real-world Deployment:** Integrating AIHire into enterprise recruitment systems requires addressing privacy, scalability, and acceptance challenges.

VIII. CONCLUSION

This paper introduced AIHire, a multi-modal framework for automated candidate evaluation. By integrating resume screening, AI-driven interviews, emotion analysis, and coding assessment, AIHire provides a holistic and fair approach to recruitment. Experimental results demonstrated improved accuracy, fairness, and efficiency compared to traditional systems. The unified candidate scorecard enhances transparency and explainability, benefiting both recruiters and candidates.

Future work will focus on expanding datasets, incorporating multilingual support, and developing recruiter dashboards. With continued refinement, AIHire has the potential to transform recruitment by reducing bias, improving scalability, and enabling explainable AI-driven hiring decisions.

REFERENCES

- [1] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proc. NAACL-HLT*, 2019.
- [2] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," in *Proc. EMNLP*, 2019.
- [3] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," OpenAI, 2022.
- [4] J. Hickman et al., "Automated Video Interview Analysis for Recruitment," *Journal of Applied Psychology*, 2022.
- [5] S. Dahabsh and R. Raut, "Resume-Job Matching using BERT and Cosine Similarity," *International Journal of Computer Applications*, 2024.
- [6] M. Muddumage et al., "Multimodal Candidate Assessment Framework," *IEEE Access*, 2022.
- [7] P. Teng and Y. Zhang, "LLM-based Coding Evaluation for Recruitment," *ACM Transactions on Computing Education*, 2024.
- [8] L. Bachman and J. Ma, "RunCodeEval: Multidimensional Coding Evaluation," *Proc. ICSE*, 2025.
- [9] H. Pata et al., "MAG-BERT-ARL: Multimodal Video Interview Analysis," *Proc. AAAI*, 2024.

- [10] S. Kumar et al., "Speech Emotion Recognition using CNN Models," *IEEE Transactions on Affective Computing*, 2023.
- [11] Y. Li et al., "PE-Net: Emotion Recognition Framework," *Pattern Recognition Letters*, 2023.